Vertexa Solutions

Response to RFP MA-RFP-2026-017

Meridian Analytics Pvt. Ltd., Hyderabad - supply of 100 business laptops

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Contact: Aditya Varma, Head of Enterprise Sales | Email: aditya@vertexasolutions.example | Phone: +91 40 5550 7707

1. Executive Summary and Understanding of the Requirement

1.1 Summary of our offer

Vertexa Solutions offers Meridian Analytics Pvt. Ltd. 100 units of the Vertexa V14 Elite, a premium 14-inch business laptop built around the Intel Core Ultra 7 155H processor, 16 GB of LPDDR5X memory, a 512 GB PCIe Gen4 SSD, a 14.0-inch 2.8K OLED display and Windows 11 Pro. The price is about Rs 62,000 per laptop. Delivery is planned in about 6 weeks from the order.

The V14 Elite is the most capable laptop in its weight class that we have built. It weighs 1.24 kg, lasts 11.2 hours in an office workload in our laboratory and keeps its performance under sustained load. The sections below set out the engineering behind those figures.

1.2 Our understanding of Meridian's requirement

Meridian is equipping new employees and has asked for 100 business laptops. They need a modern processor, at least 16 GB of memory, a 512 GB SSD, a 14-inch screen, Windows 11 Pro and a battery that lasts a full working day. We read the requirement as a request for dependable everyday machines, and we have chosen a model that gives new employees more performance than the minimum so that the laptops stay useful for the whole of their service life.

1.3 Why choose us

Vertexa Solutions is an engineering-led company. Our design centre in Pune employs 140 engineers working on hardware, firmware and thermal design, and our products are assembled at a partner factory in Chennai that follows our own test specification. We design the V14 Elite ourselves, and that is why we can describe its performance, battery behaviour and serviceability in such detail.

We care about the day-to-day experience of the person who uses the laptop. A screen that is easy on the eyes, a keyboard that is comfortable for a full day, a cooling system that is quiet and a battery that lasts are the qualities that employees notice. The V14 Elite was designed around these qualities, and the rest of this response explains how.

We also think about what happens after the first day. A laptop that stays fast after two years of updates, that keeps its battery capacity and that survives a daily commute in a backpack is a better investment than one that only looks good on a specification sheet. The V14 Elite was tested against those long-term conditions, and the results are shown in the tables in Section 2.

2. Proposed Solution (Laptop Model and Specifications)

2.1 Laptop model and specification table

Component	Vertexa V14 Elite specification
Processor	Intel Core Ultra 7 155H, 16 cores (6 performance, 8 efficient, 2 low-power efficient), 22 threads, up to 4.8 GHz turbo, 24 MB cache, 28 W sustained
Graphics and NPU	Intel Arc integrated graphics with 8 Xe cores; Intel AI Boost NPU rated at 11 TOPS
Memory	16 GB LPDDR5X-6400, dual channel, soldered
Storage	512 GB NVMe PCIe Gen4 x4 SSD in a user-replaceable M.2 2280 slot; second M.2 2242 slot free
Display	14.0-inch 2.8K (2880 x 1800) OLED, 120 Hz, 16:10, 400 nits typical and 600 nits peak in HDR, 100 percent DCI-P3, anti-reflective
Battery	75 Wh; 11.2 hours in office workload; 14.8 hours in video playback; 100 W USB-C charge to 80 percent in 45 minutes

Component	Vertexa V14 Elite specification
Weight and size	1.24 kg; 312 x 218 x 14.9 mm; magnesium-lithium alloy chassis
Ports	2 x Thunderbolt 4, 1 x USB-A 3.2 Gen 2, 1 x HDMI 2.1, 1 x microSD UHS-II reader, 1 x 3.5 mm combo audio
Wireless	Wi-Fi 7 (802.11be, 2x2) and Bluetooth 5.4
Operating system	Windows 11 Pro, 64-bit
Platform hardware	Discrete TPM 2.0 chip, fingerprint reader in the power button, 1080p IR camera with privacy shutter

2.2 Performance, battery life and build quality

The Core Ultra 7 155H combines three kinds of cores. Six performance cores handle bursts such as opening large spreadsheets, eight efficient cores handle background work, and two low-power cores handle light tasks while the screen is on and the laptop is idle. The result is a processor that finishes work quickly and then returns to a low power state, which is the main reason for the long battery life.

The integrated Arc graphics deliver about 40 percent more graphics throughput than the previous generation, and the NPU runs on-device features such as background blur and noise suppression on video calls without loading the main cores. LPDDR5X memory at 6400 MT/s provides a bandwidth of about 102 GB per second, which keeps large datasets, browser tabs and analytics tools responsive.

The chassis is a magnesium-lithium alloy that is 25 percent lighter than aluminium of the same stiffness. It has been tested with 122 cm drop tests on 6 faces and 2 edges, with a 20 kg load on the closed lid and with 30,000 hinge cycles. The result is a laptop of 14.9 mm thickness that feels solid in the hand.

2.3 Fit with Meridian's environment

The V14 Elite runs Windows 11 Pro and supports the standard device management and directory sign-in tools used in business. Thunderbolt 4 supports docks that drive two 4K displays from one cable, and the Wi-Fi 7 radio is backward compatible with Wi-Fi 6 and Wi-Fi 6E networks. Meridian's users receive a 100 W USB-C charger in the box.

The wireless design uses a Wi-Fi 7 card with two antennas placed in the display bezel, away from the metal base, which improves signal in crowded offices. Bluetooth 5.4 supports low-energy audio, so wireless headsets work for long calls with little battery drain. A fingerprint reader of the capacitive type sits inside the power button and unlocks the laptop in about 0.5 seconds.

The fingerprint reader and the IR camera both support Windows Hello, so employees can sign in with a touch or a glance. The camera shutter closes the lens physically, and a hardware switch controls the microphones.

2.4 Display

The 2.8K OLED panel is the most visible difference between the V14 Elite and a conventional business laptop. OLED pixels switch off completely when they show black, so text on a dark theme is crisp and contrast is very high. The panel covers 100 percent of the DCI-P3 colour gamut, and each unit is calibrated at the factory to an average colour error below 1.5 Delta E.

The 120 Hz refresh rate makes scrolling and cursor movement smoother, and the panel drops to 48 Hz when the content is static to save power. A low blue light mode is certified by an independent laboratory, and the panel uses DC dimming so that brightness changes cause no flicker. The 16:10 shape shows more rows of a spreadsheet than a 16:9 panel of the same width.

Audio is designed for calls and for light media use. Two 2 W woofers and two 1 W tweeters sit behind the keyboard deck and fire upward through a micro-perforated grille, and the quad-microphone array uses beamforming to pick up the speaker's voice and reduce the sound of a nearby conversation. Speech sounds clear at 70 percent volume without distortion.

2.5 Thermals, acoustics and sustained performance

Measurement	Result	Condition
Sustained processor power	28 W	30-minute all-core load, 25 degrees Celsius room
Palm-rest temperature	34 degrees Celsius	Office workload, 1 hour

Measurement	Result	Condition
Keyboard centre temperature	39 degrees Celsius	Sustained all-core load
Fan noise, office workload	31 dBA	Measured at 50 cm in a quiet room
Fan noise, full load	41 dBA	Sound level at 50 cm distance during an all-core stress run

The cooling system has two fans and a vapour chamber that spreads heat over most of the width of the laptop. The fan blades are formed from a liquid crystal polymer, which allows them to be thin and strong, and the fan curve is tuned to avoid sudden changes in sound. In office use the fans often stop completely.

We tested the cooling system in a 35 degrees Celsius chamber to simulate a warm office without air conditioning. The laptop held 24 W sustained processor power for 30 minutes, and the keyboard centre temperature reached 41 degrees Celsius. Vents are positioned on the rear edge and on the underside, so the laptop continues to cool when it sits on a soft surface for a short time.

2.6 Keyboard, touchpad and serviceability

The keyboard has 1.5 mm key travel, a spill-resistant membrane and a 3-level white backlight with an ambient light sensor. The keycaps have a slightly concave surface and a soft-touch coating that resists shine. The glass touchpad measures 120 x 75 mm and supports precision gestures.

The bottom cover is held by 8 captive Torx screws. Under it, the SSD, the battery, the Wi-Fi card and the speakers are modular, and a technician replaces any one of them in under 10 minutes. Memory is soldered to the board, which is how the design reaches its thickness and its bandwidth.

Spare parts are numbered and documented. A service guide with exploded drawings and torque values for the captive screws ships with the first order, and every replaceable part carries a part number on the part itself. A technician who removes the battery follows a four-step discharge and release process that takes about 3 minutes.

2.7 Benchmark figures

Test	Vertexa V14 Elite result	Note
Geekbench 6 single-core	2,450	Plugged in, balanced power plan
Geekbench 6 multi-core	12,900	Plugged in, balanced power plan
PCMark 10 Modern Office	7,100	On battery, balanced power plan
3DMark Time Spy graphics	3,900	Plugged in
SSD sequential read	7,000 MB per second	512 GB drive, 1 MB blocks
SSD sequential write	5,200 MB per second	512 GB drive, 1 MB blocks
Windows 11 cold boot to sign-in	11 seconds	Fast startup disabled

All figures were measured in the Vertexa laboratory in Pune on retail-configuration units with current firmware and Windows 11 Pro. Each result is the median of five runs.

2.8 Battery test results

Workload	Battery life	Test conditions
Video playback	14.8 hours	Local 1080p file, 150 nits, Wi-Fi on, volume 30 percent
Office workload	11.2 hours	Scripted document, spreadsheet and browser tasks, 150 nits
Web browsing	12.6 hours	Looped page set over Wi-Fi, 150 nits
Video conferencing	9.4 hours	Continuous call with camera on, 150 nits
Full charge time	1 hour 25 minutes	100 W adapter, laptop off

Tests began at 100 percent charge and ended at 5 percent. The results are the mean of three runs on three different units, and the spread between units was below 4 percent. The 75 Wh battery is the largest that is permitted for air travel, and it is built from cells that retain 80 percent of their capacity after 800 full cycles.

2.9 Display calibration and image quality

Display measurement	Result	Method
Colour gamut	100 percent DCI-P3	Spectroradiometer on 9 patches

Display measurement	Result	Method
Average colour error	Delta E 1.2	24-patch chart, sRGB mode
Typical brightness	400 nits	Full white field, 50 percent window
Peak HDR brightness	600 nits	10 percent window, HDR mode
Contrast ratio	Above 1,000,000 to 1	Black pixels fully off
Brightness uniformity	Within 6 percent	25-point grid at 200 nits
Pulse-width modulation	None at 60 percent brightness and above	Photodiode and oscilloscope

The calibration profile is written to each unit at the factory, and the measurement sheet is stored against the serial number. A unit that shows a colour error above Delta E 2.0 or a brightness variation above 8 percent is rejected before packing.

2.10 Camera, audio and connectivity

Feature	Detail
Camera	1080p IR camera, 90-degree field of view, Windows Hello face sign-in, physical shutter
Microphones	Quad array with beamforming and hardware mute switch
Speakers	2 x 2 W woofers and 2 x 1 W tweeters, tuned for speech clarity
Thunderbolt 4 ports	40 Gbps, DisplayPort 2.1, 100 W power delivery on both ports
Card reader	microSD UHS-II, up to 250 MB per second
Wireless antennas	4 antennas: 2 for Wi-Fi 7 and 2 for Bluetooth and sensing

Every port is tested for 10,000 insertion cycles, and the Thunderbolt ports are tested with docks from four vendors to confirm that two 4K displays run at 60 Hz from a single cable. The USB-A port supplies 1.5 A for peripherals such as headsets and wireless mouse receivers.

3. Implementation Approach, Timeline, Team and Milestones

3.1 Implementation approach

Delivery of the 100 laptops is planned in about 6 weeks from the order.

Every V14 Elite is assembled at our partner factory in Chennai on a line that follows the Vertexa production specification. Boards are tested before assembly, units are burned in for 12 hours at 40 degrees Celsius, and each finished laptop is checked against a 60-point functional list that includes display uniformity, keyboard response, camera, microphones, speakers and radios.

Packaging is designed to protect the laptop in transit. The laptop sits in a moulded pulp tray inside a rigid box, and the accessories are packed in a separate compartment. The box is made of recycled board and carries printed handling instructions. Boxes are palletised in lots of 40 and wrapped for dispatch.

Transport is by road from Chennai to our Hyderabad warehouse in sealed and tracked containers. A shock and tilt indicator is attached to each pallet, and a pallet whose indicator is triggered is opened and inspected before its boxes are released.

Firmware is controlled centrally. Every unit leaves the factory with a documented BIOS version, a controller firmware version and a driver set, and these are recorded against the serial number in our production database. A later firmware release is tested on a reference group of laptops for 2 weeks before we publish it.

The BIOS supports out-of-band updates and a rollback if an update fails to complete. A recovery partition holds a clean copy of the factory firmware, and the laptop can start from it with a key combination. These features keep the fleet at the same firmware level without a visit to each desk.

6. Support Model, Relevant Experience and References

6.1 Support model

The Vertexa V14 Elite carries a 3-year warranty.

6.2 About Vertexa Solutions

Vertexa Solutions was formed by a group of hardware engineers who wanted to build business laptops with the finish and thoughtfulness usually reserved for consumer devices. The company is headquartered in Pune, and its design centre, laboratory and product management team work from the same campus. This proximity lets an engineer walk from a thermal chamber to a design review in a few minutes.

Our product philosophy rests on three ideas. First, a laptop should disappear into the work of its user, which means low weight, low noise and long battery life. Second, a laptop should be easy to open and maintain. Third, a laptop should use materials that can be recycled when its life ends. The V14 Elite applies each of these ideas.

6.3 Materials and environment

The chassis of the V14 Elite uses 50 percent recycled magnesium alloy, and the speaker enclosures use ocean-bound recycled plastic. The packaging contains no foam and no single-use plastic. The power adapter is built from 30 percent post-consumer recycled plastic and uses a gallium nitride design that is 35 percent smaller than a silicon adapter of the same rating.

Each box also carries a short printed guide to setting up the laptop, a card that explains the 3-level backlight and the function keys, and a QR code that opens an online page of manuals and drivers.

The laptop meets the European RoHS directive on restricted substances and has been designed to comply with the e-waste rules in India. We publish the weight of each material in the product so that a recycler knows what the laptop contains. The design avoids adhesives wherever a screw can do the same work, which lets a recycler take the laptop apart quickly.

Repairability matters to us as an engineering team. A laptop that can be opened with one screwdriver and closed again without special tools extends its working life and reduces the amount of waste produced over time.

Our laboratory in Pune includes a thermal chamber, an acoustic room, a display measurement bench and a mechanical test area. Engineers use these facilities during development and again on samples from every production lot. This is why we can describe the thermal and battery behaviour of the V14 Elite with the numbers shown in Section 2.

Vertexa invests 11 percent of its yearly revenue in research and development, and about one in four of our employees works on product design. The laboratory runs a 1,000-hour reliability test on one unit from each production lot, and the results are reviewed by the product manager before the lot is released.

7. Declaration and Contact

This offer is valid for 30 days from 9 October 2026.

Authorised signatory: Aditya Varma, Head of Enterprise Sales, Vertexa Solutions.